

Foundation (Jalapeno)

- Q1.** What is the primary purpose of a box-and-whisker plot in sports statistics?
(A) To display the average score
(B) To show the distribution of scores
(C) To compare team rankings
(D) To predict game outcomes
(E) To analyze player injuries
- Q2.** The 25th percentile in a set of marathon finish times is 3 : 15. What does this represent?
(A) The fastest time
(B) The slowest time
(C) The median time
(D) The time below which 25
(D) The time above which 25
- Q3.** In a box-and-whisker plot of basketball scores, an outlier is a score that is $1.5 \cdot IQR$ above the third quartile. What does IQR represent?
(A) The difference between the highest and lowest scores
(B) The range of the middle 50
(B) The average score
(C) The median score
(D) The mode score
- Q4.** A set of data on cycling speeds has a first quartile of 15 mph and a third quartile of 25 mph. What is the interquartile range?
(A) 10 mph
(B) 15 mph
(C) 20 mph
(D) 25 mph
(E) 30 mph
- Q5.** Which of the following box-and-whisker plots indicates the presence of an outlier?
(A) A plot with a long whisker on the left side
(B) A plot with a long whisker on the right side
(C) A plot with a short box and long whiskers
(D) A plot with a long box and short whiskers
(E) A plot with no whiskers
- Q6.** A golfer's scores for 9 holes are 3, 4, 5, 4, 3, 5, 6, 4, 5. What is the second quartile (median) of these scores?
(A) 4
(B) 4.5
(C) 5
(D) 5.5
(E) 6
- Q7.** The 50th percentile in a set of swim meet times is 2 : 30. What does this represent?
(A) The fastest time
(B) The slowest time
(C) The median time
(D) The time below which 50
(D) The time above which 50
- Q8.** Which of the following statements is true about the interquartile range (IQR) in a set of data?
(A) It is the difference between the highest and lowest values
(B) It is the average of the first and third quartiles
(C) It is the range of the middle 50
(C) It is the median of the data set
(D) It is the mode of the data set

Open Ended Questions (FRQ)

- Q9.** Construct a box-and-whisker plot for the following set of data on jogging times (in minutes): 25, 30, 20, 35, 40, 28, 32, 38, 42, 45. Identify the median, first quartile, third quartile, and any outliers.
- Q10.** A set of data on tennis serve speeds has the following five-number summary: min 50 mph, Q_1 60 mph, median 65 mph, Q_3 70 mph, max 80 mph. Sketch a box-and-whisker plot for this data and explain what it reveals about the distribution of serve speeds.

Chef's Tips